



ASIC Verification
EE5213
Exercise 2 Report
FSM Reachability Analysis

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1 Exercise 2: FSM Reachability Analysis

1.1 Objective

Implement the State Reachability Analysis algorithm for an arbitrary Finite State Machine (FSM) using C++. The program computes the set of reachable states given a set of initial states and a transition relation.

1.2 Design Specifications

- **Input:** Set of initial states (I), and a transition relation (R) mapping current states to next possible states.
- **Output:** The complete set of reachable states (ARS - All Reached States).
- **Algorithm Strategy:**
 - Track **All Reached States** (ARS) and **New Next States** (NNS or Frontier Set).
 - Iteratively compute the next states (NS) from the current frontier NNS .
 - Update the frontier as newly discovered states ($NNS = NS - ARS$).
 - Add the new frontier to the all reached states ($ARS = ARS \cup NNS$).
 - Terminate when the frontier is empty.

1.3 Example FSM Analysis

To illustrate the algorithm, consider the FSM defined in the implementation with the following transition relations:

- $S_0 \rightarrow \{S_1, S_5\}$
- $S_1 \rightarrow \{S_2\}$
- $S_5 \rightarrow \{S_5, S_2\}$
- $S_2 \rightarrow \{S_3\}$
- $S_3 \rightarrow \{S_4\}$
- $S_4 \rightarrow \emptyset$

The initial state set is $I = \{S_0\}$. The state diagram is shown in Figure 1.

Step-by-Step Execution:

1. Initialization:

- $ARS = \{S_0\}$
- Frontier $NNS = \{S_0\}$

2. Iteration 1:

- Compute next states from NNS : $NS = \{S_1, S_5\}$

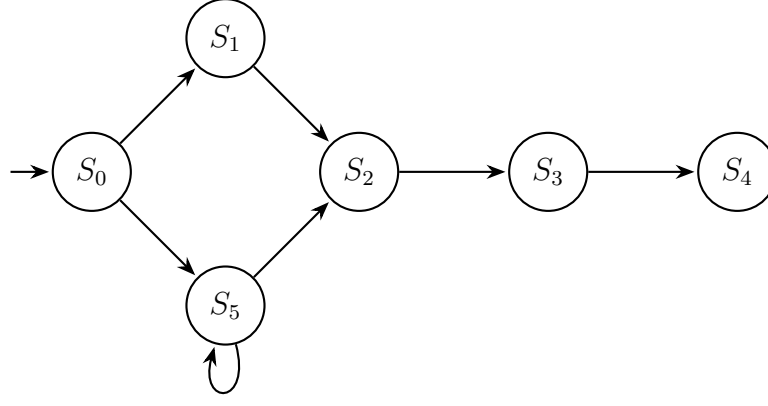


Figure 1: State Diagram of the Example FSM

- Compute new frontier: $NNS = NS \setminus ARS = \{S_1, S_5\} \setminus \{S_0\} = \{S_1, S_5\}$
- Update ARS : $ARS = ARS \cup NNS = \{S_0, S_1, S_5\}$

3. Iteration 2:

- Compute next states from NNS : $NS = \{S_2, S_5\}$ (from $S_1 \rightarrow S_2$ and $S_5 \rightarrow S_2, S_5$)
- Compute new frontier: $NNS = NS \setminus ARS = \{S_2, S_5\} \setminus \{S_0, S_1, S_5\} = \{S_2\}$
- Update ARS : $ARS = ARS \cup NNS = \{S_0, S_1, S_2, S_5\}$

4. Iteration 3:

- Compute next states from NNS : $NS = \{S_3\}$ (from $S_2 \rightarrow S_3$)
- Compute new frontier: $NNS = NS \setminus ARS = \{S_3\} \setminus \{S_0, S_1, S_2, S_5\} = \{S_3\}$
- Update ARS : $ARS = ARS \cup NNS = \{S_0, S_1, S_2, S_3, S_5\}$

5. Iteration 4:

- Compute next states from NNS : $NS = \{S_4\}$ (from $S_3 \rightarrow S_4$)
- Compute new frontier: $NNS = NS \setminus ARS = \{S_4\} \setminus \{S_0, S_1, S_2, S_3, S_5\} = \{S_4\}$
- Update ARS : $ARS = ARS \cup NNS = \{S_0, S_1, S_2, S_3, S_4, S_5\}$

6. Iteration 5:

- Compute next states from NNS : $NS = \emptyset$ (from $S_4 \rightarrow \emptyset$)
- Compute new frontier: $NNS = NS \setminus ARS = \emptyset \setminus ARS = \emptyset$
- The frontier NNS is empty, so the algorithm terminates.

Final Result: $ARS = \{S_0, S_1, S_2, S_3, S_4, S_5\}$

1.4 Implementation (fsm_reachability.cpp)

```
1 #include <iostream>
2 #include <vector>
3 #include <set>
4 #include <map>
5 #include <algorithm>
6
7 typedef int State;
8
9 void print_set(const std::string& name, const std::set<State>& s) {
10     std::cout << name << " = { ";
11     for (State val : s) std::cout << "S" << val << " ";
12     std::cout << "}\n";
13 }
14
15 std::set<State> Reachability(
16     const std::map<State, std::set<State>>& R,
17     const std::set<State>& I
18 ) {
19     std::set<State> ARS = I; // All Reached States
20     std::set<State> NNS = I; // New Next States (Frontier Set)
21     int iteration = 1;
22
23     std::cout << "--- Initialization ---\n";
24     print_set("ARS", ARS);
25     print_set("NNS", NNS);
26
27     while (true) {
28         std::cout << "\n--- Iteration " << iteration++ << " ---\n";
29         std::set<State> NS;
30         for (State s : NNS) {
31             if (R.find(s) != R.end()) {
32                 for (State next_s : R.at(s)) {
33                     NS.insert(next_s);
34                 }
35             }
36         }
37         print_set("NS ", NS);
38
39         std::set<State> new_NNS;
40         std::set_difference(
41             NS.begin(), NS.end(),
42             ARS.begin(), ARS.end(),
43             std::inserter(new_NNS, new_NNS.begin())
44         );
45         NNS = new_NNS;
46         print_set("NNS", NNS);
47
48         if (NNS.empty()) {
49             std::cout << "NNS is empty. Terminating.\n";
50             break;
51         }
52         ARS.insert(NNS.begin(), NNS.end());
53         print_set("ARS", ARS);
54     }
55     return ARS;
56 }
```

```

57
58 int main() {
59     std::map<State, std::set<State>> transition_relation = {
60         {0, {1, 5}}, {1, {2}}, {5, {5, 2}},
61         {2, {3}}, {3, {4}}, {4, {}}
62     };
63     std::set<State> initial_states = {0};
64
65     std::cout << "Starting Reachability Analysis...\n\n";
66     std::set<State> reachable_states = Reachability(transition_relation
67 , initial_states);
68
69     std::cout << "\nFinal Reachable States (ARS): { ";
70     for (State s : reachable_states) std::cout << "S" << s << " ";
71     std::cout << "}\n";
72     return 0;
73 }

```

1.5 Execution Results

The screenshot shows the OneCompiler IDE with a C++ program for Reachability Analysis. The code defines a transition relation and uses a recursive function to find reachable states. The output window shows the execution results, including the initial state, the states reached in each iteration, and the final set of reachable states (ARS).

```

1 #include <iostream>
2 #include <vector>
3 #include <set>
4 #include <map>
5 #include <algorithm>
6
7 typedef int State;
8
9 void print_set(const std::string& name, const std::set<State>& s) {
10     std::cout << name << " = { ";
11     for (State s : s) std::cout << s << " ";
12     std::cout << " }\n";
13 }
14
15 std::set<State> Reachability(
16     const std::map<State, std::set<State>&>& R,
17     const std::set<State>& S) {
18     std::set<State> ARS = S; // All Reached States
19     std::set<State> NNS = S; // New Next States (Frontier Set)
20     int iteration = 1;
21
22     std::cout << "... Initialization ... \n";
23     print_set("ARS", ARS);
24
25     while (true) {
26         std::cout << "... Iteration " << iteration << " ... \n";
27
28         std::set<State> NS;
29         for (State s : ARS) {
30             if (R.find(s) != R.end()) {
31                 for (State next_s : R.at(s)) {
32                     NS.insert(next_s);
33                 }
34             }
35         }
36         print_set("NS", NS);
37
38         std::set<State> new_NNS;
39         std::set_difference(
40             NS.begin(), NS.end(),
41             NNS.begin(), NNS.end(),
42             std::inserter(new_NNS, new_NNS.begin()));
43         NNS = new_NNS;
44         print_set("NNS", NNS);
45
46         if (NNS.empty()) {
47             std::cout << "NNS is empty. Terminating.\n";
48             break;
49         }
50         ARS.insert(NNS.begin(), NNS.end());
51         print_set("ARS", ARS);
52     }
53     return ARS;
54 }
55
56 int main() {
57     std::map<State, std::set<State>> transition_relation = {
58         {0, {1, 5}}, {1, {2}}, {5, {5, 2}},
59         {2, {3}}, {3, {4}}, {4, {}}
60     };
61     std::set<State> initial_states = {0};
62
63     std::cout << "Starting Reachability Analysis...\n\n";
64     std::set<State> reachable_states = Reachability(transition_relation, initial_states);
65
66     std::cout << "\nFinal Reachable States (ARS): { ";
67     for (State s : reachable_states) std::cout << "S" << s << " ";
68     std::cout << "}\n";
69     return 0;
70 }

```

Output:

```

Starting Reachability Analysis...

--- Initialization ---
ARS = { S0 }
NNS = { S0 }

--- Iteration 1 ---
NS = { S1 S5 }
NNS = { S1 S5 }
ARS = { S0 S1 S5 }

--- Iteration 2 ---
NS = { S2 S5 }
NNS = { S2 }
ARS = { S0 S1 S2 S5 }

--- Iteration 3 ---
NS = { S4 }
NNS = { S4 }
ARS = { S0 S1 S2 S3 S4 S5 }

--- Iteration 4 ---
NS = { S4 }
NNS = { S4 }
ARS = { S0 S1 S2 S3 S4 S5 }

--- Iteration 5 ---
NS = { }
NNS = { }
ARS is empty. Terminating.

Final Reachable States (ARS): { S0 S1 S2 S3 S4 S5 }

```

Figure 2: Program Execution Output

```

1 Starting Reachability Analysis...
2
3 --- Initialization ---
4 ARS = { S0 }
5 NNS = { S0 }
6
7 --- Iteration 1 ---
8 NS = { S1 S5 }
9 NNS = { S1 S5 }
10 ARS = { S0 S1 S5 }
11
12 --- Iteration 2 ---
13 NS = { S2 S5 }
14 NNS = { S2 }
15 ARS = { S0 S1 S2 S5 }

```

```
16
17 --- Iteration 3 ---
18 NS  = { S3 }
19 NNS = { S3 }
20 ARS = { S0 S1 S2 S3 S5 }
21
22 --- Iteration 4 ---
23 NS  = { S4 }
24 NNS = { S4 }
25 ARS = { S0 S1 S2 S3 S4 S5 }
26
27 --- Iteration 5 ---
28 NS  = { }
29 NNS = { }
30 NNS is empty. Terminating.
31
32 Final Reachable States (ARS): { S0 S1 S2 S3 S4 S5 }
```